Assignment # 01

MLOps (BS DS)

Deadline: 13-Feb-2026 @ 11:59

Total Marks = 100

Instructions:

- Combine all your work in one folder. The folder must contain only deliverable files (no binaries, no exe files, etc.).
- Rename the folder as ROLL-NUM_SECTION (e.g., 22i-0001_A) and compress the folder as a zip file. (e.g. 23i-0001_A.zip). Do not submit a .rar file.
- All the submissions will be done on Google Classroom within the deadline. Submissions other than Google Classroom (e.g., email, etc.) will not be accepted.
- The student is solely responsible for checking the final zip files for issues like corrupt files, viruses in the file, and mistakenly executed sent. If we cannot download the file from Google Classroom for any reason, it will lead to zero marks in the assignment.
- Displayed output should be well-mannered and well-presented. Use appropriate comments and indentation in your source code.
- **Be prepared for a viva or anything else after the submission of the assignment.**
- If there is a syntax error in the code, zero marks will be awarded in that part of the assignment.
- Understanding the assignment is also part of the assignment.
- **Zero marks will be awarded to the students involved in plagiarism. (Copying from the internet is the easiest way to get caught.)**
- **Late Submission** will **not** be entertained, and **no retake** request will be accepted as per the course policy.

Tip: For timely completion of the assignment, start as early as possible.

Note: Follow the given instruction; failing to do so will result in a zero.

Q#1: (100 marks)

Perform the following tasks for this question.

Task 1: (20% weightage)

- Launch an EC2 Ubuntu instance
- Attach:
  - Root volume (default)
  - One additional EBS volume (≥10 GB)
  - Format and mount the EBS volume at:
    `/mnt/ml-data`
- Ensure:
  - EBS auto-mounts after reboot
  - Training outputs are stored only on EBS

Deliverables:

- Screenshot of:
  - Mounted volume (df -h)
  - `/etc/fstab entry`
- Directory structure:

```
/mnt/ml-data/
├── datasets/
├── features/
├── models/
├── logs/
```

Task 2: (20% weightage)

- Create an S3 bucket with:
  - Versioning enabled
  - Private access only
- Upload the UCI Adult Census datasets:
  - Raw dataset (raw.csv)
  - Processed dataset (processed.csv)
- Configure AWS CLI on EC2 using IAM role (NO access keys)
- Implement:
  - One-way sync: S3 → EC2
  - One-way sync: EC2 → S3

Deliverables

- Screenshot of:
  - Bucket version history
  - Successful aws s3 sync
- Dataset stored at:
  - `/mnt/ml-data/datasets/`

Task 3: (20% weightage)

- Write a bootstrap shell script (setup_ml_env.sh) that:
  - Installs Python, pip, virtualenv
  - Creates a virtual environment

    - Installs ML libraries
- Script must be:
    - Idempotent
    - Executable on fresh EC2 launch
- Validate by:
    - Stopping EC2
    - Restarting
    - Re-running script without failure

Deliverables

- setup_ml_env.sh
- Screenshot showing:
    - Clean environment setup
    - Activated virtual environment

<u>Task 4: (25% weightage)</u>

- Build a Python ML pipeline script (NOT notebook):
    - train_pipeline.py
- Pipeline must:
    - Load data from /mnt/ml-data/datasets
    - Perform feature scaling
    - Train two ML models
    - Select best model automatically
    - Log metrics to file
- Save:
    - Best model → /mnt/ml-data/models/
    - Metrics → /mnt/ml-data/logs/

Output Example

```
Model: RandomForest
Accuracy: 0.87
Timestamp: 2026-02-04
```

<u>Task 5: (15% weightage)</u>

- Implement auto-shutdown:
    - Instance stops after inactivity OR
    - Scheduled stop using cron
- Verify:
    - Instance automatically stops
- Demonstrate:
    - Restart without data loss
    - EBS persistence

Deliverables

- Cron configuration screenshot
- EC2 stopped state evidence

- Data still available after restart

Deliverables For This Question

Students must submit:

Single ZIP file containing:

```
mlops-assignment/
├── setup_ml_env.sh
├── train_pipeline.py
├── README.md
└── screenshots/
```

README.md must include:

EC2 instance type

S3 bucket name

IAM role name

Execution steps (commands only)